

Promptathon Assignment:

Scenario 1: Search Functionality

Approach 1: Generate test cases for search feature in amazon.in

Approach 2: ICE Method

Context

Act as a Senior Test Engineer in e-commerce domain and expertise in enterprise search

Instruction

[MANDATORY] Generate testcases for search feature in amazon.in

[MANDATORY] Consider all the aspects of search functionality in a practical Indian scenario

[MANDATORY] Include Functional and Non functional(including performance) and apply testing techniques.

[MANDATORY] Include Positive and negative cases

Example:

TCID | Description | Priority | Preconditions | Steps | Expected Result

Persona

Act as a Test lead to review the testcases and provide a review score.

Output

Downloadable excel with a count of test cases

Tone:

Professional Tone with clarity

Summary of Findings – Comparison of the Two AI-Generated Responses

Executive Summary – Approach 1 Response vs. Approach 2 Response

Overall Assessment

The **Approach 2 response** demonstrates a significant improvement in **structure, scalability, and enterprise alignment** compared to the Approach 1 response. However, it still functions as a **framework/template** rather than a fully authored enterprise test suite.

Comparison Summary

Evaluation Area	Approach 1 Response	Approach 2 Response	Verdict
Functional Coverage	Good (90+ test cases)	Excellent (320 structured test cases)	Approach 2 Better
Requirement Compliance	Moderate	Excellent	Approach 2 Better
Documentation Quality	Good	Excellent (Excel, RTM, Summary)	Approach 2 Better
Enterprise Readiness	Moderate	High	Approach 2 Better
Performance Coverage	Basic	Comprehensive (Load, Stress, Spike, Soak)	Approach 2 Better
AI/ML & Search Relevance	Not covered	Included as dedicated modules	Approach 2 Better
Regional Language Support	Limited	Hindi, Tamil, Telugu, Kannada, Malayalam, Bengali, Marathi	Approach 2 Better

Strengths

Approach 1 Response

- Good functional breadth with positive and negative scenarios.
- Covers core search features (filters, sorting, pagination, accessibility, security).

- Suitable as a baseline functional test suite.

Approach 2 Response

- Well-organized Excel workbook with **320 test cases**.
- Includes **RTM, risk-based prioritization, performance workload model, AI/ML validation, personalization, regional language testing, voice/image/barcode search**.
- Better aligned with enterprise QA documentation and test management tools.

Weaknesses

Approach 1 Response

- No RTM or traceability.
- Missing priorities, testing techniques, and structured documentation.
- Limited enterprise search considerations.

Approach 2 Response

- Not yet production-ready without expanding placeholders into complete test scenarios.

Alignment with E-commerce Best Practices

Area	Approach 1	Approach 2
Functional Coverage	Good	Excellent
Error Handling	Good	Very Good
User Experience Validation	Good	Very Good
Performance Testing	Basic	Excellent
AI/ML & Personalization	Minimal	Good
Enterprise Search Validation	Limited	Good (framework level)

Final Verdict

- **Approach 1 Response:** Strong functional checklist suitable for exploratory and baseline testing

- **Approach 2 Response:** A well-structured **enterprise testing framework** with excellent coverage and documentation, but requiring detailed scenario authoring to become a production-grade test suite

Scenario 2: File Upload

Approach 1: Generate test cases for video upload feature of Youtube

Approach 2: ICE Method

Context

Act as a Senior Test Engineer in social media domain and expertise in testing social media and video based websites

Instruction

[MANDATORY] Generate testcases for video upload feature in youtube.com

[MANDATORY] Consider all the aspects of upload functionality that are possible and where the feature may go wrong.

[MANDATORY] Include Functional and Non functional(including performance) test cases and apply testing techniques that might be relevant.

[MANDATORY] Include positive,edge and negative cases including potential error conditions.

[MANDATORY] Validate before and after conditions as in pre and post upload.

Example:

TCID | Description | Priority|Preconditions| Steps | Expected Result

Persona

Act as a Test lead to review the testcases and provide a review score.

Output

Downloadable excel with a count of test cases

Tone:

Professional Tone with clarity

Summary of Findings – Comparison of the Two AI-Generated Responses

Overall Assessment

The Approach 2 response demonstrates a significant improvement in structure, scalability, and enterprise alignment compared to the Approach 1 response. However, it still functions primarily as a **test framework/template** rather than a fully authored, YouTube-specific enterprise test suite.

While the Approach 1 response provides a concise and practical set of 80 well-defined test cases covering the end-to-end upload lifecycle, the Approach 2 response expands coverage to 400 test cases and introduces enterprise testing artifacts such as severity, automation candidacy, testing techniques, preconditions, postconditions, and execution-ready documentation. However, many of the additional cases are generic template-driven scenarios rather than detailed YouTube-specific validations.

Comparison Summary

Evaluation Area	Approach 1 Response	Approach 2 Response	Verdict
Functional Coverage	Good (80 test cases)	Excellent (400 structured test cases)	Approach 2 Better
Requirement Compliance	Moderate	Excellent	Approach 2 Better
Documentation Quality	Good	Excellent (Excel, Summary, Severity, Techniques)	Approach 2 Better
Enterprise Readiness	Moderate	High	Approach 2 Better
Performance Coverage	Good	Comprehensive (Load, Scalability, Reliability, Recovery)	Approach 2 Better
Security Coverage	Good	Excellent (Security-focused modules included)	Approach 2 Better
Accessibility Coverage	Good (WCAG 2.1 scenarios)	Excellent (Dedicated Accessibility category)	Approach 2 Better

Evaluation Area	Approach 1 Response	Approach 2 Response	Verdict
Test Execution Readiness	Moderate	High	Approach 2 Better
Social Media Domain Coverage	Moderate	Moderate to Good	Slightly Better
Automation Readiness	Limited	High (Automation Candidate included)	Approach 2 Better

Strengths

Approach 1 Response

- Good functional breadth with positive, negative, and edge-case scenarios.
- Covers core upload capabilities:
 - Supported/unsupported formats
 - Metadata validation
 - Thumbnail uploads
 - Visibility settings
 - Accessibility
 - Security
 - Recovery scenarios
- Easy to review and execute manually.
- Suitable as a baseline functional test suite.

Approach 2 Response

- Well-organized Excel workbook containing 400 test cases.
- Includes:
 - Preconditions
 - Test Data
 - Detailed Steps
 - Expected Results

- Post Conditions
- Severity
- Testing Techniques
- Automation Candidacy
- Better aligned with enterprise QA repositories such as Azure DevOps, Jira/Xray, and TestRail.
- Expanded coverage for:
 - Reliability
 - Performance
 - Security
 - Accessibility
 - Compatibility
 - Localization
 - Analytics and Notifications
- Supports traceability and test management processes more effectively.

Weaknesses

Approach 1 Response

- No traceability or requirement mapping.
- Missing:
 - Preconditions
 - Detailed test steps
 - Post conditions
 - Testing techniques
 - Severity classification
 - Automation indicators
- Limited suitability for enterprise-scale execution and reporting.

Approach 2 Response

- Extensive coverage however Limited coverage of YouTube-specific business functionality such as Content ID validation, Copyright detection, Community Guidelines violations, Monetization eligibility, YouTube Shorts uploads, Live stream uploads, Creator quota limits etc.
- Not fully production-ready without converting template scenarios into detailed business-focused test cases.

Alignment with Social Media Testing Best Practices

Area	Approach 1	Approach 2
Functional Coverage	Good	Excellent
Error Handling	Good	Very Good
Upload Workflow Validation	Good	Excellent
Accessibility (WCAG)	Good	Excellent
Performance Testing	Good	Excellent
Security Testing	Good	Excellent
Recovery & Reliability	Good	Excellent
YouTube-Specific Business Rules	Limited	Moderate
Enterprise Test Management	Moderate	Excellent

Final Verdict

- **Approach 1 Response:** A strong functional checklist suitable for exploratory testing, smoke testing, and baseline validation of YouTube video upload functionality.
- **Approach 2 Response:** A well-structured enterprise testing framework with significantly improved documentation, execution readiness, and coverage.

However, it requires additional YouTube-specific business scenarios and replacement of generic template cases with fully authored test cases to become a production-grade social media test repository.

Winner: Approach 2 Response — superior in completeness, scalability, documentation quality, automation readiness, and enterprise QA applicability.

Scenario 3: API testing

Approach 1: Write API test cases for adding a new pet to the store. Refer to <https://petstore.swagger.io/#/>

Approach 2: ICE Method

Context

Act as a Senior Test Engineer with deep expertise in API testing, REST-based service validation, and test automation using the Karate DSL framework. You have hands-on experience testing public-facing APIs, validating OpenAPI/Swagger-documented services, and building maintainable, data-driven test suites for enterprise-grade applications.

Instruction

[MANDATORY] Generate comprehensive Karate DSL test cases for the "Add a New Pet" endpoint of the publicly available Pet Store API, documented at <https://petstore.swagger.io/#/> — specifically the POST /pet operation.

[MANDATORY] Before writing any test case, fetch and interpret the Swagger specification at the above URL to derive the accurate request schema, required fields, field data types, and valid enumeration values (e.g., status field: available, pending, sold).

[MANDATORY] Cover all of the following test dimensions:

- Functional — happy path, field-level validation, response schema conformance
- Negative — missing required fields, wrong data types, malformed JSON, invalid enum values
- Boundary — maximum string lengths, numeric ID limits (min/max long), empty arrays for photoUrls and tags
- Edge Cases — duplicate pet IDs, null values for optional fields, special characters in name, Unicode input, extremely large payloads
- Non-Functional — response time assertion (SLA: under 2000ms), concurrent request simulation using Karate's parallel execution

[MANDATORY]

Each test case must follow this exact tabular format:

TCID	Test Scenario	Priority	Preconditions	Karate Feature Snippet Reference	Request Payload Summary	Expected HTTP Status	Expected Response Assertions	Post-Condition
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[MANDATORY] Structure the Karate feature file output with:

- A Background section with base URL configuration and shared headers

- Individual Scenario blocks — one per test case — with descriptive names
- Inline * match assertions for response body fields: id, name, category, status, photoUrls, tags
- Response time validation using * assert responseTime < 2000
- Data-driven scenarios using Scenario Outline with Examples tables where applicable

[MANDATORY] Include positive, negative, boundary, and error-condition scenarios — minimum 20 test cases across all categories.

[MANDATORY]

Validate both pre-conditions (e.g., API is reachable, auth header if required) and post-conditions (e.g., created pet ID is retrievable via GET /pet/{petId}).

Persona

Act as a Test Lead reviewing the generated test cases for coverage completeness, assertion quality, and Karate DSL correctness. Assign a review score out of 10 per category (Functional, Negative, Boundary, Non-Functional) and an overall suite score, with a brief rationale for each score and specific improvement recommendations.

Output

Deliver the final output as a downloadable Excel file (.xlsx) containing:

- Sheet 1 — Test Cases: Full test case table with all columns as defined above, including a row count summary at the bottom
- Sheet 2 — Karate Feature Scripts: Complete feature file content per scenario, formatted for direct copy-paste into an IDE
- Sheet 3 — Review Scorecard: Test Lead review scores by category with commentary and gap analysis
- Sheet 4 — Coverage Matrix: Mapping of each test case to the API field it validates and the testing technique applied (equivalence partitioning, boundary value analysis, error guessing, etc.)

Tone

Professional, precise, and technically authoritative — written for a QA engineering audience. Comments within Karate scripts should be human-readable and self-explanatory. No ambiguity in assertions or expected outcomes.

Overall Assessment

The Approach 2 response demonstrates a significant improvement in **test coverage, enterprise readiness, traceability, and API testing alignment** compared to the Approach 1 response. However, it still functions primarily as a **structured enterprise API test inventory** rather than a fully implemented, production-ready Karate automation suite.

While the Approach 1 response provides **22 executable Karate scenarios** with reusable feature files, schema validation, and data-driven examples, the Approach 2 response expands the coverage to **40 structured API test cases** spanning functional, negative, boundary, edge-case, and non-functional testing.

It also introduces enterprise testing artifacts such as **TCIDs, priorities, preconditions, post-conditions, Karate snippet references, and execution planning**.

However, the Approach 2 response does not yet include the complete Karate DSL implementation for every test case, making it more suitable as a test design artifact than an immediately executable automation suite.

Comparison Summary

Evaluation Area	Approach 1 Response	Approach 2 Response	Verdict
Functional Coverage	Good (22 Karate scenarios)	Excellent (40 structured test cases)	Approach 2 Better
Requirement Compliance	Good	Excellent	Approach 2 Better
Documentation Quality	Good	Excellent (TCIDs, Preconditions, Post-Conditions, Summary)	Approach 2 Better
Enterprise Readiness	Moderate	High	Approach 2 Better

Evaluation Area	Approach 1 Response	Approach 2 Response	Verdict
Swagger Specification Alignment	Good	Excellent (Request schema, required fields, enums documented)	Approach 2 Better
Boundary Coverage	Moderate	Excellent (Long IDs, arrays, payload sizes, string limits)	Approach 2 Better
Negative Testing	Good	Excellent	Approach 2 Better
Edge Case Coverage	Moderate	Excellent	Approach 2 Better
Non-Functional Coverage	Basic (Response Time)	Comprehensive (Performance + Concurrent Execution Planning)	Approach 2 Better
Karate Automation Readiness	High (Executable feature scripts)	Moderate (Requires implementation in Karate)	Approach 1 Better

Strengths

Approach 1 Response

- Provides **22 executable Karate DSL scenarios** ready for automation.
- Includes:
 - Background configuration
 - Shared JSON payload
 - Schema validation
 - Scenario Outline
 - Response time assertions
 - GET verification
 - DELETE cleanup
- Good use of Karate best practices such as reusable payloads and parameterized scenarios.
- Easy to integrate into an existing Karate automation framework.

- Suitable as a baseline API automation suite.

Approach 2 Response

- Expands coverage from **22** to **40 enterprise-grade test cases**.
- Includes:
 - TCIDs
 - Priorities
 - Preconditions
 - Request payload summaries
 - Expected HTTP status
 - Response assertions
 - Post-conditions
 - Category-wise summary
- Better aligned with enterprise QA repositories such as Azure DevOps, Jira/Xray, and TestRail.
- Expanded coverage for:
 - Functional validation
 - Boundary testing
 - Edge cases
 - Negative scenarios
 - Non-functional validation
- Improves traceability and supports structured test management processes.

Weaknesses

Approach 1 Response

- Limited to **22 scenarios**, leaving several enterprise validation areas uncovered.
- Missing:
 - TCIDs
 - Priority classification

- Preconditions
- Post-conditions
- Requirement traceability
- Test management metadata
- Limited boundary and edge-case coverage.
- Less suitable for enterprise-scale reporting and governance.

Approach 2 Response

- Focuses on **test case design** rather than complete Karate DSL implementation.
- Karate Feature Snippet References are placeholders and require full scenario development.

Alignment with API Testing Best Practices

Area	Approach 1	Approach 2
Functional Coverage	Good	Excellent
Request Payload Validation	Good	Excellent
Response Assertions	Good	Excellent
Schema Validation	Excellent	Excellent
Error Handling	Good	Excellent
Boundary Value Analysis	Moderate	Excellent
Edge Case Validation	Moderate	Excellent
Performance Testing	Good	Excellent
Concurrent Execution Planning	Limited	Excellent
Swagger/OpenAPI Alignment	Good	Excellent
Enterprise Test Management	Moderate	Excellent
Karate Automation Structure	Excellent	Good (Pending full implementation)

Final Verdict

- **Approach 1 Response:** A well-structured Karate automation suite suitable for rapid implementation, proof-of-concept development, and baseline API validation. It demonstrates good adherence to Karate best practices but lacks enterprise-level documentation, traceability, and comprehensive scenario coverage.
- **Approach 2 Response:** A significantly more comprehensive enterprise API testing framework with expanded scenario coverage, improved documentation, stronger alignment with the Swagger specification, and enhanced support for QA governance and test management

Winner: Approach 2 Response

Approach 2 response is superior in terms of **coverage, enterprise documentation, traceability, boundary and edge-case validation, non-functional testing, and overall QA maturity.**

Approach 1 response remains stronger only in **immediate Karate DSL execution readiness**, whereas the Approach 2 response provides a more scalable and maintainable foundation for enterprise API testing.